

S2.2.1.

P1. Solution: Characteristic equation.

$$r^2 + r + 1 = 0$$

$$\Rightarrow r_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, r_2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

Hence,

$$e^{r_1 t} = e^{(-\frac{1}{2} + \frac{\sqrt{3}}{2}i)t} = e^{-\frac{t}{2}} \left[\cos\left(\frac{\sqrt{3}}{2}t\right) + i \sin\left(\frac{\sqrt{3}}{2}t\right) \right]$$

is a complex-valued solution.

Therefore,

$$y_1(t) = e^{-\frac{t}{2}} \cos\left(\frac{\sqrt{3}}{2}t\right)$$

$$y_2(t) = e^{-\frac{t}{2}} \sin\left(\frac{\sqrt{3}}{2}t\right)$$

are real-valued solutions. Thus

$$y(t) = e^{-t} \left[C_1 \cos\left(\frac{\sqrt{3}}{2}t\right) + C_2 \sin\left(\frac{\sqrt{3}}{2}t\right) \right]$$

P3. Solution: Characteristic equation.

$$r^2 + 2r + 3 = 0$$

$$\Rightarrow r_1 = -1 + \sqrt{2}i, r_2 = -1 - \sqrt{2}i$$

Therefore,

$$y_1(t) = e^{-t} \cos(\sqrt{2}t)$$

$$y_2(t) = e^{-t} \sin(\sqrt{2}t)$$

and then the general solution is

$$y(t) = e^{-t} \left[C_1 \cos(\sqrt{2}t) + C_2 \sin(\sqrt{2}t) \right]$$

P5. Solution: Characteristic equation.

$$r^2 + r + 2 = 0$$

$$\Rightarrow r_1 = -\frac{1}{2} + \frac{\sqrt{7}}{2}i, r_2 = -\frac{1}{2} - \frac{\sqrt{7}}{2}i$$

Therefore, the general solution is given by

$$y(t) = e^{-\frac{1}{2}t} \left[C_1 \cos\left(\frac{\sqrt{7}}{2}t\right) + C_2 \sin\left(\frac{\sqrt{7}}{2}t\right) \right]$$

$$y(0) = 1 \Rightarrow C_1 = 1$$

$$y'(0) = -2 \Rightarrow -\frac{1}{2}C_1 + \frac{\sqrt{7}}{2}C_2 = -2 \Rightarrow \begin{cases} C_1 = 1 \\ C_2 = \frac{-3}{\sqrt{7}} \end{cases}$$

$$\left(\text{Since } y'(t) = -\frac{1}{2}e^{-\frac{1}{2}t} \left[C_1 \cos\left(\frac{\sqrt{7}}{2}t\right) + C_2 \sin\left(\frac{\sqrt{7}}{2}t\right) \right] + e^{-\frac{1}{2}t} \left[-\frac{\sqrt{7}}{2}C_1 \sin\left(\frac{\sqrt{7}}{2}t\right) + \frac{\sqrt{7}}{2}C_2 \cos\left(\frac{\sqrt{7}}{2}t\right) \right] \right)$$

Thus

$$y(t) = e^{-t} \left[\cos\left(\frac{\sqrt{7}}{2}t\right) - \frac{3}{\sqrt{7}} \sin\left(\frac{\sqrt{7}}{2}t\right) \right]$$

P7. Solution: Since $b^2 - 4ac < 0$. First we know that

$$\tilde{y}_1(t) = e^{-\frac{b}{2a}t} \cos \beta t \text{ and } \tilde{y}_2(t) = e^{-\frac{b}{2a}t} \sin \beta t$$

are solutions of $\mathcal{O}$. Now,

$$\begin{aligned} y_1(t) &= e^{-\frac{b}{2a}(t-t_0)} \cos \beta(t-t_0) \\ &= e^{\frac{bt_0}{2a}} e^{-\frac{b}{2a}t} [\cos \beta t \cos \beta t_0 + \sin \beta t \sin \beta t_0] \\ &= \left(e^{\frac{bt_0}{2a}} \cos \beta t_0 \right) e^{-\frac{b}{2a}t} \cos \beta t + \left(e^{\frac{bt_0}{2a}} \sin \beta t_0 \right) e^{-\frac{b}{2a}t} \sin \beta t \\ &= C_1 \tilde{y}_1(t) + C_2 \tilde{y}_2(t) \end{aligned}$$

$$\text{where } C_1 = e^{\frac{bt_0}{2a}} \cos \beta t_0 \text{ and } C_2 = e^{\frac{bt_0}{2a}} \sin \beta t_0$$

$\Rightarrow y_1(t)$ is also a solution of $\mathcal{O}$

Similarly, we can show that $y_2(t)$ is also a solution of $\mathcal{O}$.

P8. Solution: Characteristic equation.

$$2r^2 - r + 3 = 0$$

$$\Rightarrow r_1 = \frac{1}{4} + \frac{\sqrt{23}}{4}i, r_2 = \frac{1}{4} - \frac{\sqrt{23}}{4}i$$

So that (Since $t_0 = 1$)

$$y_1(t) = e^{\frac{1}{4}(t-1)} \cos \frac{\sqrt{23}}{4}(t-1)$$

$$y_2(t) = e^{\frac{1}{4}(t-1)} \sin \frac{\sqrt{23}}{4}(t-1)$$

are real-valued solutions. Thus

$$y(t) = e^{\frac{1}{4}(t-1)} \left[C_1 \cos \frac{\sqrt{23}}{4}(t-1) + C_2 \sin \frac{\sqrt{23}}{4}(t-1) \right]$$

$$y(1) = 1 \Rightarrow C_1 = 1$$

$$y'(1) = 1 \Rightarrow \frac{1}{4}C_1 + \frac{\sqrt{23}}{4}C_2 = 1$$

$$\Rightarrow \begin{cases} C_1 = 1 \\ C_2 = \frac{3}{\sqrt{23}} \end{cases}$$

Hence,

$$y(t) = e^{\frac{1}{4}(t-1)} \left[\cos \frac{\sqrt{23}}{4}(t-1) + \frac{3}{\sqrt{23}} \sin \frac{\sqrt{23}}{4}(t-1) \right]$$